

CENTRAL LIMIT THEOREMS FOR QUADRATIC FORMS WITH TIME DOMAIN CONDITIONS ^{*†‡}

Liudas Giraitis
Boston University

Murad S. Taqqu
Boston University

July 12, 1996

Abstract

We establish the central limit theorem for quadratic forms $\sum_{t,s=1}^N b(t-s)P_{m,n}(X_t, X_s)$ of the bivariate Appell polynomials $P_{m,n}(X_t, X_s)$ under time domain conditions. These conditions relate the weights $b(t)$ and the covariances of the sequences $(P_{m,n}(X_t, X_s))$ and (X_t) . The time domain approach, together with the spectral domain approach developed earlier, yield a general set of conditions for central limit theorems.

1 Introduction

We study the central limit theorem (CLT) for quadratic forms

$$(1.1) \quad Q_N := \sum_{t,s=1}^N b(t-s)P_{m,n}(X_t, X_s)$$

where

$$P_{m,n}(X_t, X_s) =: \underbrace{X_t, \dots, X_t}_m, \underbrace{X_s, \dots, X_s}_n :$$

is a bivariate Appell polynomial (Wick power) of the linear variables X_t and X_s , $m, n \geq 0, m+n \geq 1$. Here

$$(1.2) \quad X_t = \sum_{s \in \mathbf{Z}} a(t-s)\xi_s, \quad t \in \mathbf{Z}$$

is a linear process, that is, the random variables $\xi_t, t \in \mathbf{Z}$ are independent and identically distributed, $E\xi_0 = 0$, $E\xi_0^2 = 1$, the sequence $a(t), t \in \mathbf{Z}$ of real-valued weights satisfies the condition $\sum_s a^2(s) < \infty$. We assume $E|\xi_u|^{2(m+n)} < \infty$ in order to ensure that Q_N has a

^{*}The second author was partially supported by the NSF Grants DMS-9404093 and NCR-9404931 at Boston University. The first author was supported by the NSF Grant NCR-9404931. He would like to thank Boston University for its hospitality.

[†]*AMS Subject classification.* 60F05, 62M10. *Short title:* CLT for Bivariate Appell Polynomials

[‡]*Keywords and phrases:* Appell polynomials, central limit theorem, long-range dependence, quadratic forms, time series.

finite variance. Our goal is to provide sufficient conditions for $N^{-1/2}Q_N$ to converge to a normal distribution.

The definition of multivariate Appell polynomials is given in Section 3. These polynomials are a multivariate generalization of the univariate Appell polynomials (or Hermite polynomials if (X_t) is Gaussian) and, like them, they play an important role in the limit theory of quadratic forms of dependent variables.

Central limit theorems involving Hermite or Appell polynomials have been studied by Sun [19], [20], Breuer and Major [6], Giraitis [10], Giraitis and Surgailis [11], [12], Fox and Taqqu [9], Ho and Sun [15], [16], Ho [14], Avram [2], [3], Avram and Fox [4], Giraitis and Taqqu [13], Arcones [1].

We will show, in particular, that the assumption

$$(1.3) \quad \sum_{l_1, l_2, t \in \mathbf{Z}} \left| b(l_1)b(l_2) \operatorname{Cov}(P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0, X_{l_2})) \right| < \infty,$$

which ensures that the relation $\operatorname{Var} Q_N \leq \operatorname{const} N$ holds, yields the CLT. In fact, the relation

$$\sum_{t \in \mathbf{Z}} \left| \operatorname{Cov}(P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0, X_{l_2})) \right| < \infty, \quad l_1, l_2 \in \mathbf{Z},$$

turns out to be equivalent to the condition

$$\sum_s |r(t)|^{m+n} < \infty,$$

where $r(t) = EX_t X_0$ is the covariance function of the process (X_t) and $m+n$ is the order of the Appell polynomial $P_{m,n}(X_t, X_{t+l_1})$.

The assumption (1.3) underlines the similarities between the central limit theorem for quadratic forms in linear or Gaussian variables with long-range dependence and the central limit theorem for univariate sums $S_N = N^{-1/2} \sum_{t=0}^N f(X_t)$. Breuer and Major [6] proved that if the covariance of the Gaussian stationary process (X_t) satisfies the condition $\sum_{t \in \mathbf{Z}} |r(t)|^m < \infty$ where m is the so-called ‘‘Hermite rank’’ of the function f , $Ef(X_t) = 0$ and $Ef(X_t)^2 < \infty$, then the CLT for S_N holds. Giraitis and Surgailis [11] showed that the asymptotic normality of $N^{-1/2} \sum_{t=1}^N f(X_t)$ can be established in terms of the correlation function $r_f(t) = Ef(X_t)f(X_0)$, without referring specifically to the concept of Hermite or Appell rank (it plays an important role in the proof). They showed that the CLT holds if $\sum_{t \in \mathbf{Z}} |r_f(t)| < \infty$ when X_t is Gaussian. In addition, Giraitis [10] proved that if $\sum_{t \in \mathbf{Z}} |r_f(t)| < \infty$, the CLT holds also for the linear process X_t (1.2) if the function f is sufficiently smooth, for example, if f is a polynomial.

The condition (1.3) thus extends the univariate condition $\sum_{t \in \mathbf{Z}} |r_f(t)| < \infty$ to quadratic forms. However, in contrast to the CLT for univariate sums, it does not cover certain additional cases, first discovered by Fox and Taqqu [8], where the CLT for quadratic form also holds. Specifically, it excludes the possible compensation of the long-range dependence of (X_t) by a fast decay of the weights $b(t)$. These cases, which do not have a simple formulation in the time domain, are best characterized in the spectral domain. Conditions for the CLT in these situations were obtained by Giraitis and Taqqu [13], under assumptions involving m and n and the spectral density of (X_t) .

The conditions of this paper are stated in the time domain and are, in general, not equivalent to those in Giraitis and Taqqu [13]. The method of proof for establishing the CLT is also different. Whereas in [13] we use approximation methods, here we apply the method of moments. This is because, in the spectral domain, one can approximate the possibly unbounded spectrum by a bounded one. In the time domain, however, one has to deal directly with the covariances, which can decrease slowly. The time domain conditions, expressed in terms of covariances, are easy to apply and they clarify the underlying dependence structure.

The paper is structured as follows: Section 2 contains the main results, Section 3, a description of the Appell polynomials and, Section 4, the proofs. Multivariate extensions are given in Section 5.

2 Main results

Condition (1.3) ensures the finiteness of the limiting variance and implies the CLT. This fact is stated in the following theorem.

Theorem 2.1 *Suppose*

$$(2.1) \quad \sum_{l_1, l_2, t \in \mathbf{Z}} \left| b(l_1) b(l_2) \text{Cov} (P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0, X_{l_2})) \right| < \infty.$$

If $b(0) = 0$, suppose in addition $\sum_{t \in \mathbf{Z}} |r(t)|^{m+n} < \infty$. Then the CLT holds:

$$(2.2) \quad N^{-1/2} Q_N \Rightarrow \mathcal{N}(0, \sigma^2) \quad (N \rightarrow \infty),$$

and the limiting variance is

$$(2.3) \quad \sigma^2 = \sum_{l_1, l_2, t \in \mathbf{Z}} b(l_1) b(l_2) \text{Cov} (P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0, X_{l_2})).$$

The next theorem provides a condition on r and b , under which (2.1) is satisfied.

Theorem 2.2 *If*

$$(2.4) \quad r \in L^p, \quad b \in L^q \quad (p, q \geq 1)$$

and

$$(2.5) \quad \min(mp^{-1}, 1) + \min(np^{-1}, 1) + 2q^{-1} \geq 3$$

then the CLT holds and the limiting variance is (2.3).

The next theorem makes use of Theorem 2.4 in [13] and shows that under some rather restrictive conditions on the covariance $r(t)$ and the weights $b(t)$, the long-range dependence of the (X_t) can be compensated by the fast decay of $b(t)$ in such a way that the CLT holds. These conditions ensure, in fact, that the sufficient assumptions in the spectral domain, provided in [13], are satisfied. The theorem involves *quasi-monotone* sequences: a sequence $a(t)$ is quasi-monotonically convergent to 0 if $a(t) \rightarrow 0$ and $a(t+1) \leq a(t)(1+c/t)$ as $t \rightarrow \infty$ for some $c > 0$. The sequence $a(t)$ has *bounded variation* if $\sum_{t=1}^{\infty} |a(t+1) - a(t)| < \infty$.

Theorem 2.3 Suppose $r(t) = |t|^{-\gamma_1} L_1(|t|)$, $b(t) = |t|^{-\gamma_2} L_2(|t|)$ ($0 < \gamma_1, \gamma_2 < 3$) and

$$(2.6) \quad \min(m\gamma_1, 1) + \min(n\gamma_1, 1) + 2\gamma_2 > 3,$$

where $m, n \geq 1$. Suppose in addition that both sequences $\{r(t)\}$ and $\{b(t)\}$ have bounded variation and are quasi-monotonically convergent to 0; if $1 < \gamma_1 < 3$, $r(t)$ has the same sign for large t and satisfies $\sum_{t \in \mathbf{Z}} r(t) = 0$; if $1 < \gamma_2 < 3$, $b(t)$ has the same sign for large t and satisfies $\sum_{t \in \mathbf{Z}} b(t) = 0$. Then the CLT holds. The limiting variance is expressed by (2.3) if $0 < \gamma_1, \gamma_2 < 1$.

The assumptions in Theorem 2.3 were used in Fox and Taqqu [7], Theorem 3.

Remarks.

1. Under the assumptions of the previous theorems, the finite-dimensional distributions of $\{Q_{[Nt]}/\sqrt{N}, t \geq 0\}$ converge to those of $\{\sigma B(t), t \geq 0\}$ where $B(t)$ is standard Brownian motion. The proof of this more general statement is similar to the proof of the one-dimensional theorems.
2. Condition (2.1) is not as general as the spectral domain condition (2.5) in [13] but it is formulated entirely in the time domain. Theorem 2.2 can be viewed as the time domain analogue of Theorem 2.3 of [13]. Relations (2.4), (2.5) correspond to (2.10) and (2.11) of [13] but one cannot, in general, pass from one to the other. Note also that the direction of the inequalities (2.5) here and (2.13) in [13] is reversed. Theorems 2.1, 2.2 and 2.3 are proved in Section 4. The value of a constant C that appears in the proofs may change from line to line.

3 Appell polynomials and their cumulants

Let μ be some probability measure on $\mathbf{R}$. We assume that all moments of the measure μ exists, and the mean $\int x d\mu = 0$. The (*univariate*) Appell polynomials $P_n(x)$ corresponding to the distribution μ are defined by the generating function

$$\sum_{n=0}^{\infty} \frac{z^n}{n!} P_n(x) = \frac{\exp(zx)}{\int_{\mathbf{R}} e^{zx} d\mu}.$$

They can be expressed as

$$P_n(x) = \sum_{k=0}^n \left(\sum_{\{V\}(n-k)} (-1)^r \prod_{i=1}^r \chi(|V_i|) \right) x^k$$

(see Surgailis [21]), where $\chi(k) = \chi(k, \mu)$ is the k -th cumulant of the measure μ , and the sum is taken over all partitions $(V_1, \dots, V_r)$, $r \geq 1$ of the set $\{1, \dots, n-k\}$ such that $|V_i| \geq 2$. We set $\sum_{\{V\}(0)} \dots = 1$, so that the coefficient of x^n is 1.

An important special example consists of the Hermite polynomials. These are Appell polynomials corresponding to the standard Gaussian distribution $\mu(dx) = e^{-x^2/2} dx / \sqrt{2\pi}$.

Among all Appell polynomials, the Hermite polynomials are the only one that form an orthogonal system. [18].

The Appell polynomials generalize the ordinary “powers” because they satisfy the differential equation:

$$P'_n(x) = nP_{n-1}(x)$$

with

$$EP_n(X) = \int P_n(x) d\mu(x) = 0, \quad n = 1, 2, \dots$$

providing the constants of integration. We note that

$$P_0(x) = \int d\mu = 1, \quad P_1(x) = x, \quad P_2(x) = x^2 - \int x^2 d\mu.$$

The Appell polynomials are orthogonal only when μ is a Gaussian measure. In that case, they are identical to the Hermite polynomials.

To define the multidimensional analog of the Appell polynomials it is useful to first introduce the *Wick products* (also called *Wick powers*) (cf. [12], [21]). These are multivariate polynomials:

$$: y_1, \dots, y_n :^{(\nu)} = \frac{\partial^n}{\partial z_1 \dots \partial z_n} \left[\exp\left(\sum_1^k z_j y_j\right) / \int_{\mathbf{R}^n} \exp\left(\sum_1^k z_j y_j\right) d\nu \right] \Big|_{z_1=\dots=z_n=0}$$

corresponding to a probability measure ν on R^n . If $Y_1, \dots, Y_n$ are random variables with joint distribution $\nu(dx) = P((Y_1, \dots, Y_n) \in dx)$, then

$$: Y_1, \dots, Y_n := : y_1, \dots, y_n :^{(\nu)} \Big|_{y_1=Y_1, \dots, y_n=Y_n}$$

is also called the Wick product of the random variables $Y_1, \dots, Y_n$. It is convenient to use the notation

$$: \underbrace{X_{t_1}, \dots, X_{t_1}}_{n_1}, \dots, \underbrace{X_{t_k}, \dots, X_{t_k}}_{n_k} := P_{n_1, \dots, n_k}(X_{t_1}, \dots, X_{t_k});$$

(the indices in P correspond to the number of times that the variables in “: :” are repeated). The polynomials $P_{n_1, \dots, n_k}$ can be defined also by the recurrence relations

$$\frac{\partial}{\partial x_j} P_{n_1, \dots, n_k}(x_1, \dots, x_k) = n_j P_{n_1, \dots, n_{j-1}, \dots, n_k}(x_1, \dots, x_k),$$

$$EP_{n_1, \dots, n_k}(X_{t_1}, \dots, X_{t_k}) = 0,$$

setting $P_0 \equiv 1$.

We can now relate Wick products to Appell polynomials. If P_n , $n \geq 1$ is the univariate Appell polynomial corresponding to the distribution $\mu(dx) = P(Y \in dx)$, defined earlier, then

$$: \underbrace{Y, \dots, Y}_n := P_n(Y).$$

We provide below some properties of the Wick products (cf. [12], [21]). Let W be a finite set and $Y_i, i \in W$ be a system of random variables. Let $Y^W = \prod_{i \in W} Y_i$ be the ordinary product, $: Y^W :$ the Wick product, and $\chi(Y^W) = \chi(Y_i, i \in W)$ be the cumulant of the variables $Y_i, i \in W$, respectively. We now recall the definition of the mixed cumulant:

$$\chi(Y_1, \dots, Y_n) = \frac{\partial^n}{\partial z_1 \dots \partial z_n} \log E \exp\left(\sum_{j=1}^n z_j Y_j\right) \Big|_{z_1=\dots=z_n=0}.$$

The following relations hold ([21], Prop. 1):

$$\begin{aligned} : Y^W &:= \sum_{U \subset W} Y^U \sum_{\{V\}} (-1)^r \chi(Y^{V_1}) \dots \chi(Y^{V_r}), \\ Y^W &= \sum_{U \subset W} : Y^U : \sum_{\{V\}} \chi(Y^{V_1}) \dots \chi(Y^{V_r}) = \sum_{U \subset W} : Y^U : E(Y^{W \setminus U}) \end{aligned}$$

where the sum $\sum_{U \subset W}$ is taken over all subsets $U \subset W$, including $U = \emptyset$, and the sum $\sum_{\{V\}}$ is over all partitions $\{V\} = (V_1, \dots, V_r), r \geq 1$ of the set $W \setminus U$. We define $Y^\emptyset =: Y^\emptyset := \chi(Y^\emptyset) = 1$.

It follows that Wick products are multilinear in the sense that, if $\xi_u, u \geq 1$, are independent random variables, then for any $N \geq 1$,

$$: \sum_{u_1=1}^N \lambda_{u_1}^{(1)} \xi_{u_1}, \dots, \sum_{u_n=1}^N \lambda_{u_n}^{(n)} \xi_{u_n} : = \sum_{u_1=1}^N \dots \sum_{u_n=1}^N \lambda_{u_1}^{(1)} \dots \lambda_{u_n}^{(n)} : \xi_{u_1}, \dots, \xi_{u_n} :$$

(see also Avram and Taqqu [5]). In particular, Q_N defined by (1.1), (1.2) can be written:

$$(3.1) \quad Q_N = \sum_{u_1, \dots, u_{m+n} \in \mathbf{Z}} d_N(u_1, \dots, u_{m+n}) : \xi_{u_1} \dots \xi_{u_{m+n}} :$$

where

$$(3.2) \quad d_N(u_1, \dots, u_{m+n}) = \sum_{t, s \in \mathbf{Z}} b(t-s) a(t-u_1) \dots a(t-u_m) a(s-u_{m+1}) \dots a(s-u_{m+n}).$$

Relations of this type will be used implicitly in the sequel.

An important property of the Appell polynomials is the existence of simple combinatorial rules for calculation of the (mixed) cumulants, analogous to the familiar diagrammatic formalism for the mixed cumulants of the Hermite polynomials with respect to a Gaussian measure [17]. Let us assume that W is a union of (disjoint) subsets $W_1, \dots, W_k$. If $(i, 1), (i, 2), \dots, (i, n_i)$ represent the elements of the subset $W_i, i = 1, \dots, k$, then we can represent W as a table consisting of rows $W_1, \dots, W_k$, as follows:

$$(3.3) \quad \begin{pmatrix} (1, 1), \dots, (1, n_1) \\ \dots\dots\dots \\ (k, 1), \dots, (k, n_k) \end{pmatrix} = W.$$

By a *diagram* γ we mean a partition $\gamma = (V_1, \dots, V_r), r = 1, 2, \dots$ of the table W into nonempty sets V_i (the “edges” of the diagram) such that $|V_i| \geq 1$. We shall call the edge V_i

of the diagram γ *flat*, if it is contained in one row of the table W ; and *free*, if it consists of one element, i.e. $|V_i| = 1$. We shall call the diagram *connected*, if it does not split the rows of the table W into two or more disjoint subsets. We shall call the diagram $\gamma = (V_1, \dots, V_r)$ *Gaussian*, if $|V_1| = \dots = |V_r| = 2$. Suppose given a system of random variables $Y_{i,j}$ indexed by $(i, j) \in W$. Set for $V \subset W$,

$$Y^V = \prod_{(i,j) \in V} Y_{i,j}, \quad \text{and} \quad :Y^V := (Y_{i,j}, (i,j) \in V) : .$$

For each diagram $\gamma = (V_1, \dots, V_r)$ we define the number

$$(3.4) \quad I_\gamma = \sum_{j=1}^r \chi(Y^{V_j}).$$

Proposition 3.1 (cf. [12], [21]) *Each of the numbers (i) $E(Y^{W_1} \dots Y^{W_k})$, (ii) $E(:Y^{W_1} : \dots :Y^{W_k} :)$, (iii) $\chi(Y^{W_1}, \dots, Y^{W_k})$, (iv) $\chi(:Y^{W_1} :, \dots, :Y^{W_k} :)$ is equal to*

$$\sum I_\gamma$$

where the sum is taken, respectively, over (i) all diagrams, (ii) all diagrams without flat edges, (iii) all connected diagrams, (iv) all connected diagrams without flat edges. If $EY_{i,j} = 0$ for all $(i, j) \in W$, then the diagrams in (i)-(iv) have no free edges.

We can now apply this result to linear random variables X_t , that is, of the form (1.2). The cumulant of X_t can be expressed in the form

$$\begin{aligned} \chi(X_{t_1}, \dots, X_{t_k}) &= \sum_{s_1, \dots, s_k} \chi(\xi_{s_1}, \dots, \xi_{s_k}) a(t_1 - s_1) \dots a(t_k - s_k) \\ &= \chi_k(\xi_0) \sum_s a(t_1 - s) \dots a(t_k - s), \end{aligned}$$

where $\chi_k(\xi_0) = \chi_k(\underbrace{\xi_0, \dots, \xi_0}_k)$ is the k -th cumulant of ξ_0 . This formula holds in view of the multilinearity of the cumulant and the fact that for iid variables (ξ_j) , $\chi_k(\xi_{j_1}, \dots, \xi_{j_k}) = 0$ if $j_i \neq j_l$ for some $i \neq l$. This fact, part (iv) of Proposition 3.1 and the definition (3.4) of I_γ imply the following formula for the cumulants of the Wick products of linear variables (1.2):

$$(3.5) \quad \text{cum}(:X_{1,1}, \dots, X_{1,n_1} :, \dots, :X_{k,1}, \dots, X_{k,n_k} :) = \sum_{\gamma \in \Gamma(n_1, \dots, n_k)} d_\gamma J_\gamma$$

where $\Gamma(n_1, \dots, n_k)$ denotes the set of all connected diagrams $\gamma = (V_1, \dots, V_r)$ of the table W (3.3) without flat edges, $d_\gamma = \chi_{|V_1|}(\xi_0) \dots \chi_{|V_r|}(\xi_0)$ and

(3.6)

$$J_\gamma = \sum_{s_1, \dots, s_r \in \mathbf{Z}} \prod_{j=1}^k [a(t_{j,1} - s_{j,1}) \dots a(t_{j,n_j} - s_{j,n_j}) \dots a(t_{k,1} - s_{k,1}) \dots a(t_{k,n_k} - s_{k,n_k})],$$

where $s_{i,j} \equiv s_l$ if $(i, j) \in V_l$, $l = 1, \dots, r$. We refer to (3.5) as the diagram formula.

4 Proof of the time domain theorems

We start with the following lemma which relates the covariances of the polynomials $P_{m,n}(X_t, X_{t+l})$ to those of the variables X_t .

Lemma 4.1

$$(4.1) \quad \sum_{t \in \mathbf{Z}} |\text{Cov}(P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0, X_{l_2}))| < \infty \quad \text{for any } l_1, l_2 \in \mathbf{Z}$$

if and only if

$$(4.2) \quad \sum_{t \in \mathbf{Z}} |r(t)|^{m+n} < \infty.$$

PROOF OF LEMMA 4.1. First we show that (4.1) implies (4.2). Recall that $\Gamma(m+n, m+n)$ denotes the set of connected diagrams of the table

$$W = \begin{pmatrix} (1, 1), \dots, (1, m+n) \\ (2, 1), \dots, (2, m+n) \end{pmatrix}$$

without flat edges. Here W consists of two rows and $m+n$ columns. By the diagram formula (3.5), we have

$$\begin{aligned} R(t; l_1, l_2) &:= \text{Cov}(P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0, X_{l_2})) \\ &= \sum_{\gamma=(V_1, \dots, V_r) \in \Gamma(m+n, m+n)} d(\gamma) r_{V_1}(t; l_1, l_2) \dots r_{V_r}(t; l_1, l_2) \end{aligned}$$

where

$$(4.3) \quad r_{V_l}(t; l_1, l_2) = \sum_{u \in \mathbf{Z}} \prod_{(i,j) \in V_l} a(v_{i,j} - u),$$

and $v_{i,j} = t$ if $(i, j) \in ((1, 1), \dots, (1, m))$; $v_{i,j} = t + l_1$ if $(i, j) \in ((1, m+1), \dots, (1, m+n))$; $v_{i,j} = 0$ if $(i, j) \in ((2, 1), \dots, (2, m))$; $v_{i,j} = l_2$ if $(i, j) \in ((2, m+1), \dots, (2, m+n))$. Rewrite $\Gamma(m+n, m+n)$ as the disjoint union $\Gamma_2 \cup \Gamma_{>2}$, where Γ_2 denotes the Gaussian diagrams $\gamma = (V_1, \dots, V_r)$: $|V_1| = \dots = |V_r| = 2$ and the $\Gamma_{>2}$ the diagrams γ containing $V_l \in \gamma$ such that $|V_l| > 2$. Thus,

$$(4.4) \quad R(t; l_1, l_2) = \sum_{\gamma \in \Gamma_2} + \sum_{\gamma \in \Gamma_{>2}} =: R_2(t; l_1, l_2) + R_{>2}(t; l_1, l_2).$$

In the case $l_1 = l_2 = 0$, we have

$$R_2(t; 0, 0) = \sum_{\gamma \in \Gamma_2} d(\gamma) r_{V_1}(t; 0, 0) \dots r_{V_{m+n}}(t; 0, 0) = r(t)^{m+n} \sum_{\gamma \in \Gamma_2} 1$$

since $r_V(t; 0, 0) = r(t)$ if $|V| = 2$, and $d(\gamma) = (E\xi_0^2)^{m+n} = 1$. Using (4.1), we will get

$$\left[\sum_{\gamma \in \Gamma_2} 1 \right] \sum_{t \in \mathbf{Z}} |r(t)|^{m+n} \leq \sum_{t \in \mathbf{Z}} |R(t, 0, 0)| + \sum_{t \in \mathbf{Z}} |R_{>2}(t)| < \infty,$$

provided

$$(4.5) \quad \sum_{t \in \mathbf{Z}} |R_{>2}(t; 0, 0)| < \infty.$$

We shall now prove (4.5). By definition of $\Gamma_{>2}$, there is V_i such that $|V_i| > 2$. Two cases are possible:

(a) Suppose V_i contains more than one element of both rows of the table W . Using inequalities of the form $a(t-u)a(t-v) \leq 2(a^2(t-u) + a^2(t-v))$ and (4.3), we have

$$\sum_t |r_{V_i}(t, 0, 0)| \leq 4 \sup_t |a(t)|^{|V_i|-4} \sum_{t, u \in \mathbf{Z}} a^2(t-u)a^2(-u) < \infty$$

since, $\sum_{t \in \mathbf{Z}} a^2(t) < \infty$. Therefore,

$$\sum_t |r_{V_1}(t; 0, 0)| \dots |r_{V_r}(t; 0, 0)| \leq C \sum_t |r_{V_i}(t; 0, 0)| < \infty$$

since $|r_{V_j}(t, 0, 0)| \leq C \sum_{u \in \mathbf{Z}} |a(t-u)a(-u)| \leq C \sum_{u \in \mathbf{Z}} |a(u)|^2 < \infty$ uniformly in $t \in \mathbf{Z}$ for any $V_j \in \gamma$.

(b) $|V_i| > 2$ and V_i is a “triangle” of γ , i.e. it contains only one element from the first or second row of the table W . This time, we consider $\sum r_{V_i}^2$ instead of $\sum |r_{V_i}|$. Estimating as above, we see that

$$\begin{aligned} \sum_t r_{V_i}^2(t; 0, 0) &\leq \text{const} \sum_{t \in \mathbf{Z}} \left[\sum_{u \in \mathbf{Z}} |a^2(u)a(t-u)| \right]^2 \\ &\leq \text{const} \sum_{u, u' \in \mathbf{Z}} \sum_{u \in \mathbf{Z}} a^2(u)a^2(u') \sum_{t \in \mathbf{Z}} |a(t-u)a(t-u')| \\ &\leq \text{const} \left[\sum_{u \in \mathbf{Z}} a^2(u) \right]^2 \sum_{t \in \mathbf{Z}} a^2(t) < \infty. \end{aligned}$$

In the case (b), γ contains at least two $V_i, V_j \in \gamma$ such that $|V_i| > 2, |V_j| > 2$, because W has the same number of elements in each row. Hence the Cauchy inequality yields

$$\begin{aligned} \sum_t |r_{V_1}(t, 0, 0)| \dots |r_{V_r}(t, 0, 0)| &\leq C \sum_t |r_{V_i}(t, 0, 0)| |r_{V_j}(t, 0, 0)| \\ &\leq C \left(\sum_t r_{V_i}^2(t, 0, 0) \right)^{1/2} \left(\sum_t r_{V_j}^2(t, 0, 0) \right)^{1/2} < \infty. \end{aligned}$$

Combining cases (a) and (b), we get

$$\sum_t |R_{>2}(t; 0, 0)| \leq \sum_{\gamma \in \Gamma_{>2}} |d(\gamma)| \sum_t |r_{V_1}(t, 0, 0)| \dots |r_{V_r}(t, 0, 0)| < \infty.$$

This proves (4.5) and hence (4.2). We now prove the converse, namely that (4.2) implies (4.1). Observe that in the first part of the proof, Relation (4.5) was a consequence only of $\sum_t a^2(t) < \infty$, and that, in fact, it can be strengthened to

$$\sum_{t \in \mathbf{Z}} |R_{>2}(t; l_1, l_2)| < \infty \quad \text{for any } l_1, l_2 \in \mathbf{Z}.$$

In view of (4.4), it is then sufficient to show that

$$\sum_{t \in \mathbf{Z}} |R_2(t; l_1, l_2)| < \infty.$$

But

$$\begin{aligned} \sum_{t \in \mathbf{Z}} |R_2(t; l_1, l_2)| &\leq \sum_{\gamma \in \Gamma_2} |d(\gamma)| \sum_t |r_{V_1}(t; l_1, l_2)| \dots |r_{V_{m+n}}(t; l_1, l_2)| \\ &\leq 4 \sum_{\gamma \in \Gamma_2} |d(\gamma)| \sum_t |r(t)|^{m+n} < \infty \end{aligned}$$

by (4.2), since l_1 and l_2 are fixed and

$$|r_{V_i}(t, l_1, l_2)| \leq \max \{|r(t)|, |r(t - l_2)|, |r(t + l_1)|, |r(t + l_1 - l_2)|\}.$$

□

PROOF OF THEOREM 2.1. We divide Q_N (1.1) into two parts

$$(4.6) \quad Q_N := \sum_{t,s=1}^N b(t-s) P_{m,n}(X_t, X_s) = \sum_{t,s=1: |t-s| > K}^N + \sum_{t,s=1: |t-s| \leq K}^N = Q_{N,K}^{(1)} + Q_{N,K}^{(2)}$$

where $K > 1$ is a fixed constant. The CLT in Theorem 2.1 will be proved if we show that

$$(4.7) \quad \lim_{N \rightarrow \infty} N^{-1} \text{Var } Q_{N,K}^{(1)} \leq \delta(K) \rightarrow 0 \quad (K \rightarrow \infty)$$

and

$$(4.8) \quad N^{-1/2} (Q_{N,K}^{(2)} - E Q_{N,K}^{(2)}) \Rightarrow N(0, \sigma_K^2) \quad (N \rightarrow \infty)$$

where the limit

$$(4.9) \quad \sigma_K^2 \rightarrow \sigma^2 \quad (K \rightarrow \infty)$$

exists and is defined by (2.3).

i) We first prove (4.7). Note that

$$\begin{aligned} &N^{-1} \text{Var } Q_{N,K}^{(1)} \\ &= \sum_{t,s,t',s'=1}^N b(t-s) \mathbb{I}(|t-s| > K) b(t'-s') \mathbb{I}(|t'-s'| > K) \text{Cov}(P_{m,n}(X_t, X_s), P_{m,n}(X_{t'}, X_{s'})) \end{aligned}$$

Since X is strictly stationary,

$$\text{Cov}(P_{m,n}(X_t, X_s), P_{m,n}(X_{t'}, X_{s'})) = \text{Cov}(P_{m,n}(X_{t-t'}, X_{s-t'}), P_{m,n}(X_0, X_{s'-t'})).$$

Setting $-l_1 = t - s$ and $-l_2 = t' - s'$, and denoting $t - t'$ by t , we get

$$\begin{aligned} &N^{-1} \text{Var } Q_{N,K}^{(1)} \\ &\leq \sum_{l_1, l_2, t \in \mathbf{Z}} |b(-l_1) \mathbb{I}\{|l_1| \geq K\} b(-l_2) \text{Cov}(P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0, X_{l_2}))|. \end{aligned}$$

Relation (2.1) implies that this tends to 0 as $K \rightarrow \infty$ and hence yields (4.7).

ii) We now establish the CLT (4.8) for $Q_{N,K}^{(2)}$. Assume first that all the moments of ξ_0 exit. It is then sufficient to show that all the cumulants of $N^{-1/2}Q_{N,K}^{(2)}$ of order higher than two converge to zero. We have

$$Q_{N,K}^{(2)} = \sum_{l=-K}^K b(-l) \sum_{t=1}^N P_{m,n}(X_t, X_{t+l}) + R_{N,K}$$

where the correction $R_{N,K}$ involves a finite number of terms independent of N . Since $\text{Var } R_{N,K} < \infty$ uniformly bounded in $N \geq 1$, we have $N^{-1/2}R_{N,K} \rightarrow 0$ in probability as $N \rightarrow \infty$ for any fixed $K > 0$. Set $U_N(l) = \sum_{t=1}^N P_{m,n}(X_t, X_{t+l})$. Since cumulants are multilinear, it is sufficient to show

$$(4.10) \quad \text{cum}(U_N(l_1), \dots, U_N(l_k)) = o(N^{k/2})$$

as $N \rightarrow \infty$ for any $l_1, \dots, l_k \in \mathbf{Z}$ and $k \geq 3$, and to show convergence of the covariances

$$(4.11) \quad N^{-1} \text{Cov}(U_N(l_1), U_N(l_2)) \rightarrow \sigma_{l_1, l_2} := \sum_{t \in \mathbf{Z}} \text{Cov}(P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0, X_{l_2}))$$

as $N \rightarrow \infty$ for any $l_1, l_2 \in \mathbf{Z}$.

Observe that in view of Lemma 4.1, the conditions of our theorem imply both (4.2) and (4.1). The proof of the convergence (4.10) in the case $l_1 = \dots = l_k = 0$ is given in Giraitis [10], Proposition 6, and is based on the diagram formula (3.5). In that paper it is shown that (4.2) implies the CLT for univariate Appell polynomials $P_{m+n}(X_t) := P_{m,n}(X_t, X_t)$. In the case $l_i \in \mathbf{Z}$ the cumulants of $P_{m,n}(X_t, X_{t+l_i})$ are also calculated by the diagram formula (3.5). Since the l_i 's are fixed, the proof of (4.10) turns out to be the same as in the case $l_1 = \dots = l_k = 0$.

We now turn to Relation (4.11):

$$\begin{aligned} N^{-1} \text{Cov}(U_N(l_1), U_N(l_2)) &= N^{-1} \sum_{t,s=1}^N \text{Cov}(P_{m,n}(X_{t-s}, X_{t-s+l_1}), P_{m,n}(X_0, X_{l_2})) \\ &= \sum_{v=-(N-1)}^{N-1} \left[N^{-1} \sum_{s=1 \vee (1-v)}^{N \wedge (N-v)} 1 \right] \text{Cov}(P_{m,n}(X_v, X_{v+l_1}), P_{m,n}(X_0, X_{l_2})). \end{aligned}$$

The dominated convergence theorem applies by (4.1) and hence the Relation (4.11) follows by letting $N \rightarrow \infty$. This completes the proof of the CLT (4.8) when ξ_0 has all moments.

If ξ_0 has only $2(m+n)$ moments, replace it by $\xi_0^{<a} + \xi_0^{>a}$ where $\xi_0^{<a} = \xi_0 \mathbb{I}\{|\xi_0| \leq a\} - E\xi_0 \mathbb{I}\{|\xi_0| \leq a\}$ and $\xi_0^{>a} = \xi_0 \mathbb{I}\{|\xi_0| > a\} - E\xi_0 \mathbb{I}\{|\xi_0| > a\}$. Use the multilinearity of Wick powers to decompose $Q_{N,K}^{(2)} = Q_{N,K}^{(2), <a} + Q_{N,K}^{(2), >a}$ where $Q_{N,K}^{(2), <a}$ involves only $\xi_j^{<a}$ and $Q_{N,K}^{(2), >a}$ involves at least one $\xi_j^{>a}$, i.e.

$$P_{n_1, n_2}(X_t, X_{t+l}) = \sum_{u_1, \dots, u_{m+n} \in \mathbf{Z}} a(t-u_1) \dots a(-u_m) a(t+l-u_{m+1}) \dots a(t+l-u_{m+n}) : \xi_{u_1}, \dots, \xi_{u_{m+n}} :$$

$$\begin{aligned}
&= \sum [\dots] : \xi_{u_1}^{<a}, \dots, \xi_{u_{m+n}}^{<a} : + \sum [\dots] : \xi_{u_1}, \dots, \xi_{u_{m+n}} : \mathbb{1}\{\exists s = 1, \dots, m+n : |\xi_{u_s}| > a\} \\
&=: Q_{N,K}^{(2),<a} + Q_{N,K}^{(2),>a}.
\end{aligned}$$

It is sufficient to show that

$$(4.12) \quad \limsup_a \lim_N N^{-1} \text{Var } Q_{N,K}^{(2),>a} = 0.$$

Indeed, we already know that

$$(4.13) \quad N^{-1/2} Q_{N,K}^{(2),<a} \Rightarrow N(0, \sigma_K^2(a))$$

since all moments of $Q_{N,K}^{(2),<a}$ exist. Relation (4.12) also implies that $\sigma_K^2(a) \rightarrow \sigma_K^2$ ($K \rightarrow \infty$). Then (4.8) follows from (4.12) and (4.13).

We now establish (4.12). From $\sum_{t \in \mathbf{Z}} |r(t)|^{m+n} < \infty$, using the same argument as in the proof of Lemma 4.1, we get

$$\begin{aligned}
&\sum_{t \in \mathbf{Z}} |\text{Cov}(P_{m,n}(X_t^{<a}, X_{t+l_1}^{<a}), P_{m,n}(X_0^{<a}, X_{l_2}^{<a}))| < \infty \quad \text{for any } l_1, l_2 \in \mathbf{Z}, \\
&\sum_{t \in \mathbf{Z}} |\text{Cov}(P_{m,n}(X_t, X_{t+l_1}), P_{m,n}(X_0^{<a}, X_{l_2}^{<a}))| < \infty \quad \text{for any } l_1, l_2 \in \mathbf{Z}.
\end{aligned}$$

(Only $d(\gamma)$ in Lemma 4.1 is modified.) It follows, as in the proof of (4.11), that the following limits exist:

$$\begin{aligned}
&\lim_N N^{-1} \text{Var } Q_{N,K}^{(2),<a} =: \sigma_K^{(1,1)}(a), \\
&\lim_N N^{-1} \text{Cov}(Q_{N,K}^{(2)}, Q_{N,K}^{(2),<a}) =: \sigma_K^{(1,2)}(a), \\
&\lim_N N^{-1} \text{Var } Q_{N,K}^{(2)} =: \sigma_K^2.
\end{aligned}$$

Moreover,

$$\lim_{a \rightarrow \infty} \sigma_K^{(1,1)}(a) = \lim_{a \rightarrow \infty} \sigma_K^{(1,2)}(a) = \sigma_K^2,$$

because

$$\text{cum}(\xi_{s_1}, \dots, \xi_{s_l}, \xi_{s_{l+1}}^{<a}, \dots, \xi_{s_{l+p}}^{<a}) \rightarrow \text{cum}(\xi_{s_1}, \dots, \xi_{s_{l+p}}) \text{ as } a \rightarrow \infty,$$

which equals 0 if $s_i \neq s_j$ for some $i, j = 1, \dots, l+p$ by the independence of the ξ 's. Then

$$\begin{aligned}
\lim_N N^{-1} \text{Var } Q_{N,K}^{(2),>a} &= \lim_N N^{-1} [\text{Var } Q_{N,K}^{(2)} - 2 \text{Cov}(Q_{N,K}^{(2)}, Q_{N,K}^{(2),<a}) + \text{Var } Q_{N,K}^{(2),<a}] \\
&= \sigma_K^{(1,1)}(a) - 2\sigma_K^{(1,2)}(a) + \sigma_K^2 \rightarrow 0 \quad (a \rightarrow \infty).
\end{aligned}$$

Thus, (4.12), and hence the CLT (4.8) holds.

iii) We now establish Relation (4.9). Since

$$\sigma_K^2 = N^{-1} \text{Var } Q_{N,K}^{(2)} = \sum_{l_1, l_2 = -K}^K b(-l_1) b(-l_2) N^{-1} \text{Cov}(U_N(l_1), U_N(l_2)),$$

(4.9) follows from (4.11) and (2.1). Theorem 2.1 is now proved. $\square$

Lemma 4.2 *Let $p_1 \geq 1, \dots, p_k \geq 1$, $p_1^{-1} + \dots + p_k^{-1} = k - 1$. If $b_j \in L^{p_j}$ for $j = 1, \dots, k$ then*

$$\sum_{v_1, \dots, v_{k-1} \in \mathbf{Z}} |b_1(v_1) \dots b_{k-1}(v_{k-1}) b_k(v_1 + \dots + v_{k-1})| \leq \prod_{j=1}^k \|b_j\|_{L^{p_j}}.$$

PROOF OF LEMMA 4.2. If

$$B_l(y) = \sum_{v_1, \dots, v_{l-1} \in \mathbf{Z}} |b_1(v_1) \dots b_{l-1}(v_{l-1}) b_l(v_1 + \dots + v_{l-1} + y)|$$

then, after a change of variables,

$$\sum_{v_1, \dots, v_{k-1} \in \mathbf{Z}} |b_1(v_1) \dots b_{k-1}(v_{k-1}) b_k(v_1 + \dots + v_{k-1})| = \sum_v b_k(v) B_{k-1}(v) \leq \|b_k\|_{L^{p_k}} \|B_{k-1}\|_{L^{r_{k-1}}}$$

where $p_k^{-1} + r_k^{-1} = 1$ by Hölder inequality. We shall now use the Young's inequality

$$\|f * g\|_{L^r} \leq \|f\|_{L^p} \|g\|_{L^q}, \quad 1 + r^{-1} = p^{-1} + q^{-1}, \quad p, q, r \geq 1.$$

Applying it $k - 1$ times gives

$$\|B_{k-1}\|_{L^{r_{k-1}}} \leq \|b_{k-1}\|_{L^{p_{k-1}}} \|B_{k-2}\|_{L^{r_{k-2}}} \leq \dots \leq \prod_{j=1}^k \|b_j\|_{L^{p_j}}$$

where $1 + r_{k-1}^{-1} = p_{k-1}^{-1} + r_{k-2}^{-1}, \dots, 1 + r_2^{-1} = p_2^{-1} + p_1^{-1}$. Summing these relations and relation $1 = p_k^{-1} + r_k^{-1}$ gives $k - 1 = p_1^{-1} + \dots + p_k^{-1}$. (Observe also that $r_j \geq 1$, $j = 1, \dots, k$.) $\square$

PROOF OF THEOREM 2.2. As in proof of Theorem 2.1 it is sufficient to show that the relations (4.7)-(4.9) are satisfied. Note, that $\sum_{t \in \mathbf{Z}} |r(t)|^{m+n} < \infty$, since (2.4) and (2.5) imply that $m + n \geq p$. Therefore, (4.8) holds by the same arguments as in the proof of Theorem 2.1. Let us now verify (4.7). Observe that if

$$S := \sum_{u_1, \dots, u_k \in \mathbf{Z}} c(u_1, \dots, u_k) : \xi_{u_1} \dots \xi_{u_k} :$$

with weights $c(u_1, \dots, u_k)$ satisfying the condition $c(\cdot) \in L^2(Z^k)$, then

$$\text{Var } S = ES^2 = \sum_{\gamma \in \Gamma(k, k)} \sum_{u_1, \dots, u_r \in \mathbf{Z}} c(u_{11}, \dots, u_{1k}) c(u_{21}, \dots, u_{2k}) d(\gamma)$$

by the diagram formula (3.5). Since the first sum involves only finitely many diagrams, we can use the Cauchy formula to get

$$\text{Var } S \leq C(k, \xi_0) \sum_{u_1, \dots, u_k \in \mathbf{Z}} c^2(u_1, \dots, u_k)$$

where the constant $C(k, \xi_0)$ depends only on k and on the distribution of ξ_0 . Therefore, for $Q_{N,K}^{(1)}$ defined as (4.6), we get

$$\text{Var } Q_{N,K}^{(1)} \leq C(\xi_0) \sum_{t_1, s_1, t_2, s_2=1}^N |b(t_1 - s_1) \mathbb{I}\{|t_1 - s_1| \geq K\} b(t_2 - s_2) r^m(t_1 - t_2) r^n(s_1 - s_2)|$$

$$(4.14) \quad \leq C(\xi_0)N \sum_{v_1, v_2, v_3 \in \mathbf{Z}} |b(v_1)1\{|v_1| \geq K\}b(v_2)r^m(v_3)r^n(v_1 + v_2 + v_3)| =: NV_K.$$

Let $m, n \geq 1$. By assumption, $p_1^{-1} + p_2^{-1} + 2q^{-1} \geq 3$ where $p_1^{-1} = \min(mp^{-1}, 1)$ and $p_2^{-1} = \min(np^{-1}, 1)$. (Since for sequences, $r \in L^p \Rightarrow r \in L^{p'}$ with $p' \geq p$, we can assume $p_1^{-1} + p_2^{-1} + 2q^{-1} = 3$, by increasing for example p_1). We can now apply the Lemma 4.2 to get

$$V_K \leq C(\xi_0) \|b \mathbb{1}(|v| \geq K)\|_{L^q} \|b\|_{L^q} \|r^m\|_{L^{p_1}} \|r^n\|_{L^{p_2}} \rightarrow 0 \quad (K \rightarrow \infty)$$

since $\|r^m\|_{L^{p_1}} \leq (\sum |r(t)|^{m(\max(m^{-1}p, 1))^{1/p_1}} \leq (\sum |r(t)|^{\max(p, m)})^{1/p_1} \leq C(\sum |r(t)|^p)^{1/p_1} < \infty$, since $r \in L^p$. Similarly $\|r^n\|_{L^{p_2}} < \infty$.

Assume now that $m \geq 1, n = 0$. Then the condition (2.5), namely, $\min(mp^{-1}, 1) + 2q^{-1} \geq 3$ implies that $q = 1$ and $m \geq p$. Therefore, again $V_K \rightarrow 0$ ($K \rightarrow \infty$) since both b and r^m in (4.14) are in L^1 . This establishes (4.7).

It remains to verify Relation (4.9), namely, that σ_K^2 converges to σ^2 , given by (2.3), as $K \rightarrow \infty$. Since $Q_{N,K}^{(1)} + Q_{N,K}^{(2)} = Q_N$ and $\text{Var } Q_{N,K}^{(2)} = \text{Var } Q_{N,K}^{(1)} - 2\text{Cov}(Q_N, Q_{N,K}^{(1)}) + \text{Var } Q_{N,K}^{(1)}$, then

$$\begin{aligned} |\sigma_K^2 - \sigma_{K'}^2| &\leq \limsup_N N^{-1} |\text{Var } Q_{N,K}^{(2)} - \text{Var } Q_{N,K'}^{(2)}| \\ &\leq N^{-1} (2[\text{Var } Q_N \text{Var } Q_{N,K}^{(1)}]^{1/2} + 2[\text{Var } Q_N \text{Var } Q_{N,K'}^{(1)}]^{1/2} + \text{Var } Q_{N,K}^{(1)} + \text{Var } Q_{N,K'}^{(1)}). \end{aligned}$$

Now, $\text{Var } Q_{N,K}^{(1)} \leq CNV_K$ by (4.14), $\text{Var } Q_{N,K}^{(2)} \leq CN$ since $Q_{N,K}^{(2)}$ satisfies the CLT (Relation (4.8)), and hence $\text{Var } Q_N \leq 2\text{Var } Q_{N,K}^{(1)} + 2\text{Var } Q_{N,K}^{(2)} \leq CN$ as $N \rightarrow \infty$. Since $V_K \rightarrow 0$ as $K \rightarrow \infty$, we get

$$|\sigma_K^2 - \sigma_{K'}^2| \rightarrow 0 \quad (K, K' \rightarrow \infty).$$

Thus $\sigma_K^2 \rightarrow \sigma^2 < \infty$ and hence (4.9) holds and the sum on the right hand side of (2.3) converges. This establishes the CLT with limiting variance σ^2 given by (2.3). $\square$

PROOF OF THEOREM 2.3. The assumptions imply that f and $\hat{b}$ satisfy the assumptions $f(x) \leq C|x|^{-\alpha}$, $\hat{b}(x) \leq C|x|^{-\beta}$, $x \in [-\pi, \pi]$ of Theorem 2.4 in [13] with the $\alpha = 1 - \gamma_1$ and $\beta = 1 - \gamma_2$. (For details see the proof of Fox and Taqqu [7], Theorem 3). Moreover, if $0 < \gamma_1, \gamma_2 < 1$, then $r \in L^p$, $b \in L^q$ with $p > [\gamma_1]^{-1}, q > [\gamma_2]^{-1}$. Then the assumptions (2.4), (2.5) of Theorem 2.2 are satisfied as well and hence the limiting variance can also be represented in the form (2.3). $\square$

5 Multivariate generalizations

The following result is a multivariate generalization of Theorem 2.2. Let

$$X_t^{(i,l)} = \sum_{u \in \mathbf{Z}} a^{(i,l)}(t-u) \xi_u, \quad t \in \mathbf{Z}, \quad i = 1, \dots, k; \quad l = 1, 2$$

be $2k$ stationary time series with same mean 0, finite variance innovations sequence $\{\xi_u, u \in \mathbf{Z}\}$ and consider

$$Q_N^{(i)} := \sum_{t,s=1}^N b_i(t-s) P_{m_i, n_i}(X_t^{(i,1)}, X_s^{(i,2)}),$$

$i = 1, \dots, k$. Assume $E|\xi_0|^{2(m_i+n_i)} < \infty$, in order to ensure that $Q_N^{(i)}$ has finite second moments for all i . Let $r_{(i,l),(i',l')}(t) = EX_t^{(i,l)}X_0^{(i',l')}$, $1 \leq i, i' \leq k, 1 \leq l, l' \leq 2$, denote the cross-covariances.

Theorem 5.1 *Suppose that each quadratic form*

$$Q_N^{(i)} := \sum_{t,s=1}^N b_i(t-s)P_{m_i,n_i}(X_t^{(i,1)}, X_s^{(i,2)}), \quad i = 1, \dots, k$$

satisfies the assumptions

$$(5.1) \sum_{l_1, l_2, t \in \mathbf{Z}} \left| b_i(l_1)b_j(l_2) \text{Cov}(P_{m_i,n_i}(X_t^{(i,l)}, X_{t+l_1}^{(i,2)}), P_{m_j,n_j}(X_0^{(j,l)}, X_{l_2}^{(j,2)})) \right| < \infty, \quad 1 \leq i, j \leq k$$

and

$$(5.2) \sum_{t \in \mathbf{Z}} |r_{(i,l),(i',l')}(t)|^{m_i+n_i} < \infty, \quad 1 \leq i, i' \leq k, 1 \leq l, l' \leq 2.$$

Then as $N \rightarrow \infty$

$$(5.3) \quad N^{-1/2}(Q_N^{(1)}, \dots, Q_N^{(k)}) \Rightarrow (Z^{(1)}, \dots, Z^{(k)}),$$

where $(Z^{(1)}, \dots, Z^{(k)})$ is the Gaussian vector with zero mean and cross-covariances

$$(5.4)$$

$$\sigma_{i,j} \equiv EZ^{(i)}Z^{(j)} := \sum_{l_1, l_2, t \in \mathbf{Z}} b_i(l_1)b_j(l_2) \text{Cov}(P_{m_i,n_i}(X_t^{(i,1)}, X_{t+l_1}^{(i,2)}), P_{m_j,n_j}(X_0^{(j,1)}, X_{l_2}^{(j,2)})).$$

Remark. In view of Lemma 4.1 Condition (5.2) is automatically satisfied for any $i = i'$ if $X^{(i,1)} = X^{(i,2)}$ and $b_i(0) \neq 0$.

PROOF OF THEOREM 5.1. It is sufficient to show that

$$(5.5) \quad S_N = N^{-1/2} \sum_{i=1}^k c_i Q_N^{(i)} \Rightarrow S := \sum_{i=1}^k c_i Z^{(i)} \quad (N \rightarrow \infty)$$

for any real numbers c_i , $i = 1, \dots, k$. As in the proof of Theorem 2.1 we divide $Q_N^{(i)}$ (1.1) into two parts:

$$Q_N^{(i)} := \sum_{t,s=1}^N b_i(t-s)P_{m,n}(X_t^{(i,1)}, X_s^{(i,2)}) = \sum_{t,s=1:|t-s|>K}^N + \sum_{t,s=1:|t-s|\leq K}^N = Q_{N,K}^{(i,1)} + Q_{N,K}^{(i,2)},$$

where $K > 1$ is a fixed constant. Then the multivariate CLT (5.3) follows from the relations:

$$(5.6) \quad \lim_{N \rightarrow \infty} N^{-1} \text{Var} Q_{N,K}^{(i,1)} \leq \delta(K) \rightarrow 0 \quad (K \rightarrow \infty) \quad i = 1, \dots, k;$$

$$(5.7) \quad N^{-1/2} \sum_{i=1}^k c_i Q_{N,K}^{(i,2)} \Rightarrow \sum_{i=1}^k c_i Z_K^{(i)} \quad (N \rightarrow \infty);$$

$$(5.8) \quad EZ_K^{(i)} Z_K^{(j)} \rightarrow EZ^{(i)} Z^{(j)} \quad (K \rightarrow \infty), \quad 1 \leq i, j \leq k.$$

Since the cumulants are multilinear, the proof of the relations (5.6), (5.7) and (5.8) under assumptions (5.1) and (5.2), can be obtained in the same way as that of the relations (4.7), (4.8) and (4.9) in the proof of Theorem 2.1, respectively. $\square$

Theorem 5.2 *The statement of the Theorem 5.1 remains true if Condition (5.1) is replaced by:*

$$(5.9) \quad r_{(i,1),(i,1)} \in L^{p(i,1)}, \quad r_{(i,2),(i,2)} \in L^{p(i,2)}, \quad b_i \in L^{q_i} \quad (p_{(i,1)}, p_{(i,2)}, q_i \geq 1),$$

for $1 \leq i \leq k$, and

$$(5.10) \quad \min(m_i p_{(i,1)}^{-1}, 1) + \min(n_i p_{(i,2)}^{-1}, 1) + 2q_i^{-1} \geq 3 \quad (1 \leq i \leq k).$$

In addition, $a^{(i,1)} \in L^1$ if $k > 1$, $m_i = 1, n_i = 0$, and $a^{(i,2)} \in L^1$ if $k > 1$, $m_i = 0, n_i = 1$.

Remark. This extra condition was not required in the “univariate” version, Theorem 2.2, where only one form was considered.

PROOF OF THEOREM 5.2. One has to prove (5.6), (5.7) and (5.8). An argument similar to that of Theorem 2.2 shows that under the assumptions (5.9) and (5.10) the relation (5.6) is valid. It remains to check the limits (5.7) and (5.8).

Since the proof of (5.8) is similar to that of (4.9) in Theorem 2.2, we only show here that (5.7) holds.

Set $U_N^{(i)}(l) = \sum_{t=1}^N P_{m_i, n_i}(X_t^{(i,1)}, X_{t+l}^{(i,2)})$, $i = 1, \dots, k$. Since cumulants are multilinear, it is sufficient to show, as in the proof of Theorem 2.2, that

$$(5.11) \quad \text{cum}(U_N^{(i_1)}(l_1), \dots, U_N^{(i_p)}(l_p)) = o(N^{p/2})$$

as $N \rightarrow \infty$ for any $l_1, \dots, l_p \in \mathbf{Z}$, $1 \leq i_1, \dots, i_p \leq k$ and $p \geq 3$, and to prove convergence of the cross-covariances

$$(5.12) \quad N^{-1} \text{Cov}(U_N^{(i_1)}(l_1), U_N^{(i_2)}(l_2)) \rightarrow \sigma_{l_1, l_2} := \sum_{t \in \mathbf{Z}} \text{Cov}(P_{m,n}(X_t^{(i_1,1)}, X_{t+l_1}^{(i_1,2)}), P_{m,n}(X_0^{(i_2,1)}, X_{l_2}^{(i_2,2)}))$$

as $N \rightarrow \infty$ for any $l_1, l_2 \in \mathbf{Z}$ and $1 \leq i_1, i_2 \leq k$. The convergence of the cumulants (5.11) can be obtained similarly as Relation (4.10) in Theorem 2.2, and its proof is based on the assumption (5.2).

The proof of (5.12) is more involved. Clearly, it is sufficient to check that

$$(5.13) \quad \sum_{t \in \mathbf{Z}} |R(t; l_1, l_2)| < \infty$$

where

$$\begin{aligned} R(t; l_1, l_2) &:= \text{Cov}(P_{m,n}(X_t^{(i_1,1)}, X_{t+l_1}^{(i_1,2)}), P_{m,n}(X_0^{(i_2,1)}, X_{l_2}^{(i_2,2)})) \\ &= \sum_{\gamma=(V_1, \dots, V_r) \in \Gamma(m_{i_1}+n_{i_1}, m_{i_2}+n_{i_2})} d(\gamma) r_{V_1}(t) \dots r_{V_r}(t). \end{aligned}$$

Here

$$r_V(t) \equiv r_V(t; l_1, l_2) = \sum_{u \in \mathbf{Z}} a^{(i_1, 1)}(t-u)^{n_1(V)} a^{(i_1, 2)}(t+l_1-u)^{n_2(V)} a^{(i_2, 1)}(-u)^{n_3(V)} a^{(i_2, 2)}(l_2-u)^{n_4(V)}$$

and $n_1(V) = |V \cap \{(1, 1), \dots, (1, m_{i_1})\}|$, $n_2(V) = |V \cap \{(1, m_{i_1} + 1), \dots, (1, m_{i_1} + n_{i_1})\}|$, $n_3(V) = |V \cap \{(2, 1), \dots, (2, m_{i_2})\}|$, $n_4(V) = |V \cap \{(2, m_{i_2} + 1), \dots, (2, m_{i_2} + n_{i_2})\}|$. Relation (5.13) will follow if we show that

$$(5.14) \quad \sum_{t \in \mathbf{Z}} |r_{V_1}(t) \dots r_{V_r}(t)| < \infty$$

for any $\gamma \in \Gamma(m_{i_1} + n_{i_1}, m_{i_2} + n_{i_2})$. Since $|V_1| + \dots + |V_r| \equiv |W| = m_{i_1} + n_{i_1} + m_{i_2} + n_{i_2}$ then by Hölder inequality

$$(5.15) \quad \sum_t |r_{V_1}(t) \dots r_{V_r}(t)| \leq \prod_{i=1}^r \left(\sum_t |r_{V_i}(t)|^{\frac{|W|}{|V_i|}} \right)^{\frac{|V_i|}{|W|}}$$

Set $q_i = \sum_t |r_{V_i}(t)|^{\frac{|W|}{|V_i|}}$, $i = 1, \dots, r$.

If $|V_i| = 2$, ("Gaussian" V_i), namely, $V_i = ((i_1, l), (i_2, l'))$, then $|W|/|V_i| = |W|/2 \geq \min(m_{i_1} + n_{i_1}, m_{i_2} + n_{i_2})$, and therefore, $q_i \leq C \sum_t |r_{(i_1, l), (i_2, l')}(t)|^{\min(m_{i_1} + n_{i_1}, m_{i_2} + n_{i_2})} < \infty$ by assumption (5.2).

Recall that V_i is a "triangle" if $|V_i| > 2$ and if it contains only one element from the first or second row of the table W . If V_i is non-Gaussian non-"triangle", then as in the proof of case (a), Lemma 4.1, we have that $q_i < \infty$. Thus, if $\gamma = (V_1, \dots, V_r)$ consists of either Gaussian or contains non-Gaussian non-"triangle" V_i 's, then (5.14) holds.

Suppose now that γ contains only one "triangle" V_i . (To simplify the notations, set $i = 1$ and suppose that it is the first row of the table W that contains a single element of the "triangle" V_1). In contrast to the case (b) of Lemma 4.1, we have to consider three scenarios:

(b1) Suppose all other V_j are "Gaussian", $j = 2, \dots, r$ and $r \geq 2$. Then $r = m_{i_1} + n_{i_1}$ and there are $m_{i_1} + n_{i_1} - 1$ Gaussian V_j 's. Since $r \geq 2$, by Cauchy inequality,

$$\begin{aligned} \sum_{t \in \mathbf{Z}} |r_{V_1}(t) \dots r_{V_r}(t)| &\leq C \left(\sum_{t \in \mathbf{Z}} |r_{V_1}(t)|^2 \right)^{1/2} \left(\sum_{t \in \mathbf{Z}} |r_{V_2}(t) \dots r_{V_r}(t)|^2 \right)^{1/2} \\ &\leq C \left(\sum_{t \in \mathbf{Z}} |r_{V_1}(t)|^2 \right)^{1/2} \max_{j=2, \dots, r} \left(\sum_{t \in \mathbf{Z}} |r_{V_j}(t)|^{2(m_{i_1} + n_{i_1} - 1)} \right)^{1/2} < \infty \end{aligned}$$

since $\sum_{t \in \mathbf{Z}} |r_{V_1}(t)|^2 < \infty$ for the "triangle" V_1 (see case (b) in the proof of Lemma 4.1), and $\sum_{t \in \mathbf{Z}} |r_{V_j}(t)|^{2(m_{i_1} + n_{i_1} - 1)} < \infty$ for the $j = 2, \dots, r$ "Gaussian" V_j 's. Indeed, $2(m_{i_1} + n_{i_1} - 1) \geq m_{i_1} + n_{i_1}$ and for Gaussian V_j , r_{V_j} is a cross-covariance, so we can use Assumption (5.2).

(b2) Suppose $r = 1$, i.e. γ consists of a single "triangle" V_1 . (Assume without loss of generality $m_{i_1} = 1, n_{i_1} = 0$.) Then

$$q_1 = \sum_{t \in \mathbf{Z}} |r_{V_1}(t)| \leq C \sum_{t, u \in \mathbf{Z}} |a^{(i_1, 1)}(t-u)| \max(|a^{(i_2, 1)}(u)|^2, |a^{(i_2, 2)}(u)|^2) < \infty$$

using the special assumption $a^{(i_1,1)} \in L^1$ which holds in this case, and the standard assumptions $a^{(i_1,1)}, a^{(i_1,2)} \in L^2$.

(b3) Suppose that γ contains more than one "triangle" (say V_1, V_2). Then

$$\sum_{t \in \mathbf{Z}} |r_{V_1}(t) \cdots r_{V_r}(t)| \leq C \sum_{t \in \mathbf{Z}} |r_{V_1}(t) r_{V_2}(t)| \leq C \left(\sum_{t \in \mathbf{Z}} |r_{V_1}(t)|^2 \right)^{1/2} \left(\sum_{t \in \mathbf{Z}} |r_{V_2}(t)|^2 \right)^{1/2} < \infty.$$

Relation (5.13) is now proved. □

In the next result the weights are different, but the processes are identical.

Corollary 5.1 *Suppose that*

$$Q_N^{(i)} := \sum_{t,s=1}^N b_i(t-s) P_{m_i, n_i}(X_t, X_s),$$

$i = 1, \dots, k$, where $X(t) = \sum_{u \in \mathbf{Z}} a(t-u) \xi_u$, $t \in \mathbf{Z}$ is a linear process.

A1) *If $m_i + n_i = 1$ for some $i = 1, \dots, k$, assume:*

$$a \in L^1 \quad \text{and} \quad \begin{cases} b_j \in L^1 & \text{for all } j=1, \dots, k \quad \text{such that } m_i = 0 \text{ or } n_i = 0 \\ b_j \in L^2 & \text{for all } j=1, \dots, k \quad \text{such that } m_j \geq 1 \text{ and } n_j \geq 1. \end{cases}$$

A2) *If $m_i + n_i \geq 1$ for all $i = 1, \dots, k$, assume*

$$r \in L^p, \quad b_i \in L^{q_i} \quad (q_i \geq 1, i = 1, \dots, k),$$

and

$$\min(m_i p^{-1}, 1) + \min(n_i p^{-1}, 1) + 2q_i^{-1} \geq 3 \quad (i = 1, \dots, k).$$

Then the multivariate CLT (5.3) holds.

PROOF OF COROLLARY 5.1. Follows from Theorem 5.2. □

References

- [1] M. A. Arcones. Limit theorems for non-linear functionals of a stationary Gaussian sequence of vectors. *The Annals of Probability*, 22:2242–2274, 1994.
- [2] F. Avram. On bilinear forms in Gaussian random variables and Toeplitz matrices. *Probability Theory and Related Fields*, 79(1):37–46, 1988.
- [3] F. Avram. Generalized Szegő theorems and asymptotics of cumulants by graphical methods. *Transactions of the American Mathematical Society*, 330(2):637–649, 1992.
- [4] F. Avram and R. Fox. Central limit theorems for sums of Wick products of stationary sequences. *Transactions of the American Mathematical Society*, 330(2):651–663, 1992.
- [5] F. Avram and M. S. Taqqu. Noncentral limit theorems and Appell polynomials. *The Annals of Probability*, 15:767–775, 1987.

- [6] P. Breuer and P. Major. Central limit theorems for non-linear functionals of Gaussian fields. *Journal of Multivariate Analysis*, 13:425–441, 1983.
- [7] R. Fox and M. S. Taqqu. Non-central limit theorems for quadratic forms in random variables having long-range dependence. *The Annals of Probability*, 13:428–446, 1985.
- [8] R. Fox and M. S. Taqqu. Large-sample properties of parameter estimates for strongly dependent stationary Gaussian time series. *The Annals of Statistics*, 14:517–532, 1986.
- [9] R. Fox and M. S. Taqqu. Central limit theorems for quadratic forms in random variables having long-range dependence. *Probability Theory and Related Fields*, 74:213–240, 1987.
- [10] L. Giraitis. Central limit theorem for functionals of a linear process. *Lithuanian Mathematical Journal*, 25:25–35, 1985.
- [11] L. Giraitis and D. Surgailis. CLT and other limit theorems for functionals of Gaussian processes. *Probability Theory and Related Fields*, 70:191–212, 1985.
- [12] L. Giraitis and D. Surgailis. Multivariate Appell polynomials and the central limit theorem. In E. Eberlein and M. S. Taqqu, editors, *Dependence in Probability and Statistics*. Birkhauser, New York, 1986.
- [13] L. Giraitis and M. S. Taqqu. Limit theorem for bivariate Appell polynomials: Part I. Central limit theorems. *Probability Theory and Related Fields*, 1996. To appear.
- [14] H. C. Ho. On limiting distributions of nonlinear functions of noisy Gaussian sequences. *Stochastic Analysis and Applications*, 10:417–430, 1992.
- [15] H. C. Ho and T. C. Sun. A central limit theorem for non-instantaneous filters of a stationary Gaussian process. *Journal of Multivariate Analysis*, 22:144–155, 1987.
- [16] H. C. Ho and T. C. Sun. Limiting distributions of nonlinear vector functions of stationary Gaussian processes. *The Annals of Probability*, 18:1159–1173, 1990.
- [17] V. A. Malyshev. Cluster expansions in lattice models of statistical physics and the quantum theory of fields. *Russian Mathematical Surveys*, 35(2):1–62, 1980.
- [18] J. Shohat. The relation of the classical orthogonal polynomials to the polynomials of Appell. *American Journal of Math.*, 58:453–464, 1936.
- [19] T. C. Sun. A central limit theorem for non-linear functions of a normal stationary process. *Journal of Mathematics and Mechanics*, 12:945–978, 1963.
- [20] T. C. Sun. Some further results on central limit theorems for non-linear functions of a normal stationary process. *Journal of Mathematics and Mechanics*, 14:71–85, 1965.
- [21] D. Surgailis. On Poisson multiple stochastic integral and associated equilibrium Markov process. In *Theory and Applications of Random Fields*, pages 233–238. Springer-Verlag, Berlin, 1983. In: *Lecture Notes in Control and Information Science*, Vol. **49**.

Liudas Giraitis, Boston University, Department of Mathematics, 111 Cummington Street, Boston, MA 02215, USA. *email*: liudas@math.bu.edu

Murad S. Taqqu, Boston University, Department of Mathematics, 111 Cummington Street, Boston, MA 02215, USA. *email*: murad@math.bu.edu